Business Problem About Retail Relabeling Using Logistic Regression

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Part 1 - Feature Engineering and EDA

1.

```
# Create competitor average
df["CompAvg"] = df[
    ["Competitor_BigBoxDepot",
     "Competitor_ValueMart",
     "Competitor_QuickShop"]
].mean(axis=1)

# Create price gap variables
df["GapPct"] = 100 * (df["StorePrice"] - df["CompAvg"]) / df["CompAvg"]
df["AbsGapPct"] = abs(df["GapPct"])

# Create log units
df["LogUnits"] = np.log(df["WeeklyUnitsSold"])

# Create margin percentage
df["MarginPct"] = 100 * (df["StorePrice"] - df["UnitCost"]) / df["StorePrice"]
```

2. Report the relabel base …

```
Overall relabel rate:
0.4

Relabel rate by Category:
Category
Apparel         0.390244
Electronics     0.439791
Grocery         0.411523
Household       0.378378
PersonalCare    0.383333
Name: Relabel, dtype: float64

Relabel rate by Electronic Shelf Label:
ElectronicShelfLabel
0    0.338756
1    0.540659
Name: Relabel, dtype: float64
```

Interpret the ESL split in terms of menu costs…

ESL items had a higher relabeling rate (54.07%) than non-ESL items (33.88%). This is consistent with the menu-cost hypothesis: because ESL makes price changes much cheaper and easier, the store can reprice items more frequently. Paper tags have higher labor and material costs, creating a greater deterrent to relabeling.

Part 2 - A naive model

3. Fit Model 1

```
Optimization terminated successfully.
         Current function value: 0.621007
         Iterations 5

MODEL 1 RESULTS
===================
                           Logit Regression Results
==============================================================================
Dep. Variable:                Relabel   No. Observations:                 1500
Model:                          Logit   Df Residuals:                     1496
Method:                           MLE   Df Model:                            3
Date:                Sun, 27 Sep 2026   Pseudo R-squ.:                0.07727
Time:                        18:36:28   Log-Likelihood:                -931.51
converged:                       True   LL-Null:                       -1009.5
Covariance Type:            nonrobust   LLR p-value:                 1.328e-33
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.0589      0.098      0.603      0.547      -0.133       0.250
GapPct         0.0379      0.005      8.383      0.000       0.029       0.047
UnitCost       0.0005      0.003      0.154      0.877      -0.006       0.007
RelabelCost   -0.6387      0.071     -9.031      0.000      -0.777      -0.500
==============================================================================

Pseudo R-squared: 0.0772720635772195
AIC: 1871.020000316912
```

4. Explain deductively why the signed gap is the wrong functional form for this decision: what does an owner do about an item that is 10% under competitors versus 10% over?

The signed gap is the wrong functional form because both directions of mispricing can lead to a price change. An item 10% below competitors may need a price increase, while an item 10% above competitors may need a price decrease. Both represent a 10% misalignment, so the model should use AbsGapPct rather than the signed GapPct.

Part 3 - The economically motivated model

5. Fit Model 2

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                Relabel   No. Observations:                 1500
Model:                          Logit   Df Residuals:                     1493
Method:                           MLE   Df Model:                            6
Date:                Sun, 27 Sep 2026   Pseudo R-squ.:                  0.2750
Time:                        18:36:28   Log-Likelihood:                -731.91
converged:                       True   LL-Null:                       -1009.5
Covariance Type:            nonrobust   LLR p-value:                1.059e-116
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -4.2772      0.567     -7.540      0.000      -5.389      -3.165
AbsGapPct      0.1778      0.011     16.245      0.000       0.156       0.199
CostChangePct  0.1276      0.017      7.584      0.000       0.095       0.161
LogUnits       0.4324      0.131      3.300      0.001       0.176       0.689
RelabelCost   -0.8589      0.085    -10.085      0.000      -1.026      -0.692
DaysSinceLastChange  0.0040      0.001      6.369      0.000       0.003       0.005
UnitCost       0.0004      0.004      0.102      0.919      -0.007       0.008
==============================================================================
```

6. Odds ratios with 95% confidence intervals

Variable	Odds Ratio	95% CI	p-value
AbsGapPct	1.195	1.169 - 1.221	<0.001
CostChangePct	1.136	1.099 - 1.174	<0.001
LogUnits	1.541	1.192 - 1.992	0.001
RelabelCost	0.424	0.359 - 0.501	<0.001
DaysSinceLastChange	1.004	1.003 - 1.005	<0.001
UnitCost	1.000	0.993 - 1.008	0.919

Holding other variables fixed, each additional percentage point of AbsGapPct is associated with 19.46% higher odds of relabeling. This is an odds change, not a percentage-point probability increase.

Each additional $1 RelabelCost is associated with 57.64% lower odds of relabeling, holding other variables fixed. Higher menu costs discourage marginal corrections.

Doubling weekly sales is associated with about 34.95% higher relabel odds: $\exp(0.4324 \times \ln 2) \approx 1.3495$. A fixed tag cost can be recovered more quickly on faster-selling items.

7. Why CostChangePct is significant but UnitCost is not

A supplier cost increase can squeeze an existing margin and trigger repricing; CostChangePct is strongly significant ($p < 0.001$). UnitCost alone does not indicate that a price is wrong: an expensive item may already be priced appropriately. Its $p = 0.919$ and interval spanning 1 provide no evidence of an independent level effect here.

8. Model comparison

Metric	Model 1	Model 2
Pseudo R^2	0.077	0.275
AIC	1871.0	1477.8
BIC	1892.3	1515.0

Higher pseudo-R^2 and lower AIC/BIC are preferred. Model 2 raises pseudo-R^2 from 0.077 to 0.275 while reducing both penalized fit criteria substantially. We would deploy its specification in a monitored pilot, using held-out performance and net benefit to guide operations.

Part 4 - Predictive evaluation

70/30 stratified split, random_state = 7, threshold = 0.5. Test set N = 450.

	Predicted 0	**Predicted 1**
Actual 0	233	37
Actual 1	64	116

Accuracy = 0.776; AUC = 0.853

Accuracy exceeds the 60.0% majority-class baseline. There are 37 false positives and 64 false negatives, so the model misses more observed changes than it falsely flags. AUC measures ranking across thresholds: an observed relabel receives a higher score than a non-relabel about 85.33% of the time. These are predictions of historical behavior, not proof of optimal decisions.

Part 5 - Economic decision analysis
Memo:

To: Store Owner
From: Group 3
Subject: Economic Decision Analysis

After reviewing the pricing data for 1,500 products, we found that most products would benefit from a pricing review. The analysis showed that approximately 98.8% of SKUs had a positive net benefit. In simple terms, this means the estimated money being lost because prices are not aligned with the market is greater than the cost of updating the shelf labels. Based on these results, reviewing and adjusting prices would be financially worthwhile for most products in the store.

Now to understand what drives pricing decisions, we used a logistic regression model that evaluated several factors related to relabeling. The final model was selected because it provided a stronger overall fit and better explained relabeling behavior than the simpler model. The results showed that products with larger price differences from competitors were more likely to require a pricing review. Supplier cost increases were also an important factor, suggesting that rising costs often create pressure to update pricing in order to protect profit margins. Higher-selling products were more likely to be relabeled because even small pricing issues can have a larger financial impact when many units are sold. Products that had gone longer without a price update were also more likely to need review. In contrast, higher relabeling costs made price changes less likely because management must balance the benefit of repricing against the cost of updating shelf labels. Interestingly, the actual cost of the product itself was not an important factor. This suggests that changes in supplier costs matter more than the product's cost level when pricing decisions are made.

These findings suggest that high-selling products with greater price differences from rivals, recent increases in supplier costs, and longer intervals between price updates ought to be given priority when it comes to pricing reviews.

The results also support expanding the use of Electronic Shelf Labels (ESLs). Since ESLs make price updates faster and less expensive, they allow the store to respond more quickly to supplier cost increases and competitor price changes. Relabeling rates were 54.07% for ESL products and 33.88% for non-ESL products, a difference of around 20.2 percentage points. This can improve pricing accuracy, reduce labor costs, and support more effective pricing decisions over time. Additionally, the projected benefit of correcting pricing misalignment surpasses the estimated cost of relabeling for the majority of SKUs, as indicated by the roughly 98.8% positive net benefit rate.

One important limitation is that this analysis is based on historical store data rather than a controlled experiment. Because of this, the results identify patterns and relationships in the data, but they do not prove that one factor directly causes another. Consequently, the estimated effects are associational rather than causal. Even so, the findings provide useful guidance for identifying which products should be repriced first and where ESL investments may provide the greatest benefit.

Canvas Questions:

Q1. Model 1 vs. Model 2

A signed GapPct treats underpricing and overpricing as opposite signals even though both can motivate relabeling, diluting a real misalignment effect. Model 2 raises pseudo R^2 from 0.0773 to 0.2750 and lowers AIC from 1,871.020 to 1,477.817. However, it also adds cost changes, sales volume, and staleness, so the improvement cannot be attributed solely to replacing GapPct with AbsGapPct.

Q2. MarginPct

MarginPct is worth testing as a candidate predictor, but it should not be added to the deployed model until it is validated. MarginPct could capture the current margin cushion and help explain the urgency of cost increases or the room for price reductions. It overlaps with StorePrice and UnitCost, potentially adding redundancy. Verify that prices precede the decision to avoid outcome leakage, then compare an expanded model through training-set cross-validation. The notebook does not test this extension, so no improvement is established.

Q3. In-sample fit vs. out-of-sample generalization

Model 2 was the better choice because it included additional factors that help explain why products are relabeled, including AbsGapPct, CostChangePct, LogUnits, RelabelCost,DaysSinceLastChange, and UnitCost. The model did a better job of explaining relabeling decisions within the dataset and provided a stronger overall fit than Model 1. While it is important for a model to fit the data it was built on, it also needs to perform well when making predictions on new data. This is known as out-of-sample generalization. A model that performs well on both current and future data is more useful for business decision-making. Based on the results, Model 2 provided a better balance between explaining past relabeling decisions and supporting future pricing decisions.

Q4. From statistical significance to business decisions

A simple way to interpret the results is to look at the p-values. As a general rule, a p-value below 0.05 means the factor is important and is likely related to whether a product gets relabeled.

In Model 2, AbsGapPct ($p < 0.001$) was important, which means products with bigger pricing differences compared to competitors were more likely to need a price review. CostChangePct ($p < 0.001$) was also important, showing that when supplier costs increase, the store is more likely to update prices to protect profit margins. LogUnits ($p = 0.001$) was significant as well, meaning higher-selling products receive more pricing attention because they have a larger impact on revenue.

The model also showed that RelabelCost ($p < 0.001$) was important. In simple terms, the more expensive it is to change a shelf label, the less likely a price change becomes. DaysSinceLastChange ($p < 0.001$) was another important factor, suggesting that products that have not been reviewed in a long time are more likely to need a price update.

Finally, UnitCost ($p = 0.919$) was not significant because its p-value was much higher than 0.05. This suggests that the actual cost of a product is not as important as changes in supplier costs when it comes to deciding whether a product should be repriced.

From a business perspective, these results help identify where management should focus its efforts. Products with large pricing gaps, increasing supplier costs, high sales volume, and older prices should be reviewed first. The Net Benefit analysis also showed that about 98.8% of products would generate more value from correcting pricing differences than it would cost to update the shelf labels, making repricing worthwhile for most products in the dataset.